U-A-500B
March 30, 1976
SUPERSEDING
Int. Fed. Spec. U-A-00500a(DSA-DM)
July 24, 1968, and
Fed. Spec. U-A-500
September 23, 1959

FEDERAL SPECIFICATION

AMMONIA, AROMATIC INHALANTS

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers ammonia inhalant solution, aromatic, in ampuls (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-676	- Boxes, Setup.
PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

Federal Standards:

Fed. Std. No. 123	- Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

FSC 6505

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. Department of Health, Education, and Welfare:

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, U.S. Department of Health, Education, and Welfare, Washington, DC 20204.)

U.S. Pharmacopeial Convention, Inc.:

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

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3. REQUIREMENTS

3.1 Material. Shall be aromatic ammonia solution, in ampuls. The material shall be an aromatic ammonia inhalant solution complying with all requirements in this specification. A coloring agent may be added provided it is harmless and does not hamper the efficacy of the solution.1/ The ammonia solution and the ethyl alcohol used in the manufacture of the final solution shall be in accordance with the tests, standards, and requirements of the U.S.P. for "Ammonia Water, Stronger" and "Alcohol", respectively. A suitably denatured ethyl alcohol containing no volatile denaturants may be used in lieu of Ethyl Alcohol, U.S.P. The formulation of the denatured ethyl alcohol shall be such that it would be acceptable to the Federal Food and Drug Administration (F.D.A.). All other ingredients entering into the final solution shall be of U.S.P. or N.F. quality, or if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.1.1 Total ammonia content. The total ammonia content shall be not less than 15.0 percent weight/weight (w/w) and not more than 22.0 percent w/w when determined according to the method specified in 4.5.1.

3.1.2 Ethyl alcohol content. The ethyl alcohol content shall be not less than 35.0 percent volume/volume (v/v) and not more than 40.0 percent v/v when determined according to the method specified in 4.5.2.

3.2 Immediate container. Immediate container for the aromatic ammonia solution shall be an ampul in conformance with the following requirements:

3.2.1 Ampul. Ampul shall be a glass tube having overall dimensions as follows: Diameter of not less than 7/32 inch and not more than 10/32 inch, length of not less than 1 inch and not more than 1-7/16 inch, and a wall thickness of 0.15 to 0.30 millimeter. Ampul shall be hermetically heat sealed at both ends.

3.2.2 Contents. Each ampul shall contain not less than 1/3 ml of aromatic ammonia solution when tested as determined in 4.5.3.

3.2.3 Compatibility. The immediate container shall be acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

1/ If coloring agents are used in the solution, such coloring agent(s) shall be certified or listed as harmless and suitable for coloring drugs under the terms of the Federal Food, Drug, and Cosmetic Act and the Color Additive Regulations.

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3.2.4 Cleanliness. The immediate container shall be free from dirt, foreign matter, or other contaminants, both before and after filling.

3.2.5 Pressure. Ampuls shall not leak or break and shall be capable of withstanding in a low-pressure chamber, a pressure of 1.68 pounds per square inch (p.s.i.) (equivalent to subjecting the ampuls to an altitude of 50,000 feet). When placed in the covering specified in 3.2.6, the ampul shall be capable of being crushed easily between thumb and forefinger.

3.2.6 Covering.

3.2.6.1 Inner covering. Ampul shall be covered with clean, white, absorbent cotton or suitable cotton cloth which shall be of sufficient thickness to prevent penetration of crushed glass. The cotton cloth shall provide for the proper distribution of the contents and shall prevent any dripping of aromatic ammonia solution when the ampul is crushed.

3.2.6.2 Outher covering. The absorbent cotton or suitable cotton shall be covered by one of the following:

(a) The use of a woven mesh netting of artificial silk, rayon, or mercerized cotton which shall not fray at the ends. The netting shall be securely fastened or otherwise attached to prevent unraveling around each cotton or cloth covered ampul.

(b) The use of a starched, white (bleached) absorbent cotton which shall be securely fastened or otherwise attached to prevent raveling.

(c) The use of a non-woven, rayon-polypropylene carded wrapping material with side seam and end seams heat-sealed to prevent accidental opening.

3.2.7 Overall dimensions. Overall dimensions of the immediate container (ampul) (with the inner and outer coverings in place) shall not exceed 2 inches in length and shall not exceed 3/8 inch in diameter.

3.3 Delivery schedule. Not more than 1 year shall have elapsed from the date of manufacture of the product, to the date of delivery to the Government.

3.3.1 Date of manufacture. The date of manufacture shall be indicated by month and year, such as "MFD 6/68", or month, date, and year, (such as: "MFD June 24, 1968"). The date of manufacture shall be the date when the solution is placed within the immediate container (ampul).

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3.4 Workmanship. All materials used in the manufacture of the material and its containers shall be first class throughout, and shall be free from any defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of aromatic ammonia solution produced from one formulation, subjected to the same compounding operation and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling

For visual examination	Inspection level	AQL (defects per 100 units)	Unit of product
Major A	II (normal, single)	1.0	Covered ampul
Minor A	S-2	2.5	Covered ampul
Minor B	S-2	2.5	Ampul

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4.3.1 End item testing. For end item testing, acceptance number shall be zero (0) for all sample sizes.

TABLE II. End item testing

Characteristic	Requirement paragraph	Test paragraph	Appendix level	Minimum sample
Ammonia assay	3.1.1	4.5.1	S-2 1/	10
Alcohol content	3.1.2	4.5.2	S-4 1/	75
Contents	3.2.2	4.5.3	S-3	32
Pressure	3.2.5	---	S-2	35

1/ Composite for each test.

4.4 Examination. The aromatic ammonia solution shall be examined to determine compliance with all requirements contained in this specification.

4.4.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects

Categories	Defects 1/
Major A	
101	Ampul not fully covered with absorbent cotton or suitable cotton cloth.
102	Outher covering not adequately secured.
103	Aromatic ammonia solution and immediate container not free of interaction.
104	Immediate container not free of dirt, foreign matter, or other contaminants.
Minor A	
201	Covered ampul does not crush easily under finger pressure.
202	Covered ampul overall dimensions not as specified.
Minor B	
251	Ampul overall dimensions not as specified.
252	Ampul wall thickness not as specified.
253	Ampul not filled (non-fill or leaker).

1/ Examination is not restricted to the classified possible defects listed above.

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4.5 Tests.

4.5.1 Total ammonia determination. Randomly select ten unbroken ampuls, from which the inner and outer coverings have been removed, weigh the ten ampuls accurately, and carefully place them in a 500 milliliter (ml) Erlenmeyer flask containing exactly 200.0 ml of 0.5N sulfuric acid. Carefully crush the ampuls in the acid solution and stir the mixture until complete reaction between the ammonia and the acid has taken place. Titrate the excess sulfuric acid with 0.5N sulfuric acid using methyl red indicator. Each ml of 0.5N sulfuric acid is equivalent to 0.008516 grams of NH_3. Filter the broken glass from the ampuls through a Gooch crucible. Wash with purified water, and follow with acetone. Dry to constant weight at 100°C ± 2°C, cool, and weigh. Calculate the weight of glass. The weight of the sample (aromatic ammonia solution) is determined by difference.

Calculation:

$$\frac{\text{ml of 0.5N } H_2SO_4 \text{ consumed X 0.008516 X 100}}{\text{Weight of sample (Gram)}} = \text{Percent of ammonia w/w.}$$

4.5.2 Ethyl alcohol determination. Ethyl alcohol shall be determined in accordance with the general procedure specified in the U.S.P. under "Alcohol Determination."

4.5.3 Contents. Carefully remove the outer and inner coverings of the ampuls. Break each ampul separately in a small beaker (50 ml) with a suitable rod. Immediately pour the solution into a graduated cylinder calibrated to at least .02 ml. Alternates: (a) the quantity of solution in each ampul may be measured by drawing the solution from the beaker, by suitable means, into a measuring pipet calibrated to at least .02 ml; (b) the quantity of solution in each ampul may be measured by drawing the solution directly into an accurately calibrated hypodermic syringe.

4.6 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with acceptable quality level of 4.0 expressed in terms of percent defective.

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TABLE IV. Examination of preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.2.1 Unit of issue (for military use only). One package containing ten (10) ampuls, as specified, constitues one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. Immediate containers shall be as specified in 3.2.

5.2.2.2 Unit package. Each unit (10 ampuls) shall be packaged in a box constructed in accordance with PPP-B-566 or PPP-B-676. Dimensions of box shall be not more than 4 inches long, 5/8 inches wide, and 2-1/8 inches in depth. Closure of box shall be adequate to prevent accidental opening.

5.2.2.3 Intermediate package. Ten (10) units shall be packaged as specified for procedure code 2 in PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

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5.3 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.3.1 Level A. Four hundred eighty (480) units shall be packed as specified for procedure code 2 in PPP-C-186.

5.3.2 Level B. Four hundred eighty (480) units shall be packed as specified for procedure code 2 in PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Civil agencies. The unit intermediate and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military activities.

5.4.2.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below.

5.4.2.2 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

5.4.2.3 Unit package. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.
(b) the national stock number.
(c) the date of manufacture prefixed by "MFD".

Notes: (1) The national stock number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

5.4.2.4 Intermediate package. Each intermediate package shall be marked as specified in PPP-C-186. Date of manufacture shall be shown in lieu of date packed.

5.4.2.5 Exterior container. Exterior container shall be marked as specified in PPP-C-186. Date of manufacture shall be shown in lieu of date packed.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6505-00-106-0875	AMMONIA INHALANT SOLUTION, AROMATIC, Ampuls, 1/3 ml, 10s

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

MILITARY CUSTODIANS:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Civil Agency Coordinating Activities:
GSA - FSS
HEW - FDA, NIH
VA - DMS

Project No. 6505-1400

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